LAPORAN OPTIMASI DATABASE — MINGGU 5

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1. Data Pengujian

Dataset dibuat dengan `generate_performance_data` dan `generate_relationship_data` (repeatable, dijalankan dua kali tanpa menambah data). Lecturer/Student/Lesson/Enrollment/Assignment/Submission di bawah sudah mencakup data seed Minggu 4 (+5 course, +1 dosen, +1 mahasiswa, 3 lesson, 1 enrollment, 2 assignment, 2 submission).

Tabel	Jumlah
Course (total)	10.005
Course PERF	10.000
Lecturer	6
Student	51
Lesson	503
Enrollment	1.001
Assignment	202
Submission	1.002

Lingkungan: PostgreSQL 16.15 (Ubuntu 16.15-0ubuntu0.24.04.1), Django 5.2, dijalankan di sandbox pengerjaan (bukan di Docker Compose). Waktu = median dari 7 pengulangan setelah 1x warm-up.

2. Hasil Pengujian

Checkpoint 1 — Baseline query

```
Jumlah Course: 10005
Jumlah query SQL: 1
```

Tabel perbandingan BEFORE vs AFTER

Percobaan	Query BEFORE	Query AFTER	Waktu BEFORE	Waktu AFTER	Teknik
Course + Lecturer (100 course)	101	1	54.782 ms	2.632 ms	select_related()
Course + Lesson (50 course)	51	2	42.319 ms	6.074 ms	prefetch_related()

Database Index	EXPLAIN ANALYZE BEFORE (tanpa index)	EXPLAIN ANALYZE AFTER (dengan index)	Execution time
Query modul: is_active=True, title='Performance Course 5000' (0 baris)	Seq Scan, 10.005 baris dibuang oleh filter	Index Scan using catalog_cou _is_acti_d7f8f8_idx	0.603 ms → 0.018 ms
Judul aktif: title='Performance Course 5001' (1 baris)	Seq Scan, 10.004 baris dibuang oleh filter	Index Scan using catalog_cou _is_acti_d7f8f8_idx	0.568 ms → 0.022 ms

Catatan: pada modul, query index memakai judul *Performance Course 5000*. Karena generator membuat course dengan `i % 5 == 0` menjadi tidak aktif, course 5000 berstatus `is_active=False` sehingga query tersebut mengembalikan 0 baris. Karena itu ditambahkan pengujian kedua dengan judul aktif (5001) yang mengembalikan 1 baris.

EXPLAIN ANALYZE — Query modul (judul 5000, is_active=True)

BEFORE (index dilepas sementara):

```
Sort  (cost=239.07..239.08 rows=1 width=58) (actual time=0.594..0.595 rows=0 loops=1)
  Sort Key: code
  Sort Method: quicksort  Memory: 25kB
  ->  Seq Scan on catalog_course  (cost=0.00..239.06 rows=1 width=58) (actual time=0.592..0.592
    rows=0 loops=1)
        Filter: (is_active AND ((title)::text = 'Performance Course 5000'::text))
        Rows Removed by Filter: 10005
Planning Time: 0.041 ms
Execution Time: 0.603 ms
```

AFTER (index dipasang):

```
Sort  (cost=8.31..8.32 rows=1 width=58) (actual time=0.009..0.009 rows=0 loops=1)
  Sort Key: code
  Sort Method: quicksort  Memory: 25kB
  ->  Index Scan using catalog_cou_is_acti_d7f8f8_idx on catalog_course  (cost=0.29..8.30 rows=1
    width=58) (actual time=0.007..0.007 rows=0 loops=1)
        Index Cond: ((is_active = true) AND ((title)::text = 'Performance Course 5000'::text))
Planning Time: 0.046 ms
Execution Time: 0.018 ms
```

EXPLAIN ANALYZE — Query judul aktif (judul 5001, is_active=True)

BEFORE (index dilepas sementara):

```
Sort  (cost=239.07..239.08 rows=1 width=58) (actual time=0.540..0.541 rows=1 loops=1)
  Sort Key: code
  Sort Method: quicksort  Memory: 25kB
  ->  Seq Scan on catalog_course  (cost=0.00..239.06 rows=1 width=58) (actual time=0.267..0.537
    rows=1 loops=1)
        Filter: (is_active AND ((title)::text = 'Performance Course 5001'::text))
        Rows Removed by Filter: 10004
Planning Time: 0.037 ms
Execution Time: 0.549 ms
```

AFTER (index dipasang):

```
Sort  (cost=8.31..8.32 rows=1 width=58) (actual time=0.013..0.013 rows=1 loops=1)
  Sort Key: code
  Sort Method: quicksort  Memory: 25kB
  ->  Index Scan using catalog_cou_is_acti_d7f8f8_idx on catalog_course  (cost=0.29..8.30 rows=1
    width=58) (actual time=0.009..0.010 rows=1 loops=1)
        Index Cond: ((is_active = true) AND ((title)::text = 'Performance Course 5001'::text))
Planning Time: 0.044 ms
Execution Time: 0.022 ms
```

EXPLAIN code='PERF05000' (index unique bawaan field code)

```
Index Scan using catalog_course_code_e109f879_like on catalog_course  (cost=0.29..8.30 rows=1
    width=58) (actual time=0.006..0.006 rows=1 loops=1)
  Index Cond: ((code)::text = 'PERF05000'::text)
Planning Time: 0.028 ms
Execution Time: 0.013 ms
```

Lab D — exists(), count(), values(), annotate()

```
exists: True
count: 10000
[{'code': 'PERF00000', 'title': 'Performance Course 0'}, {'code': 'PERF00001', 'title': 'Performance
    Course 1'}, {'code': 'PERF00002', 'title': 'Performance Course 2'}]
PERF00000 10
PERF00001 10
PERF00002 10
```

Pengujian aplikasi (python manage.py test catalog)

```
  Applying contenttypes.0001_initial... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0001_initial... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
  Applying auth.0004_alter_user_username_opts... OK
  Applying auth.0005_alter_user_last_login_null... OK
  Applying auth.0006_require_contenttypes_0002... OK
  Applying auth.0007_alter_validators_add_error_messages... OK
  Applying auth.0008_alter_user_username_max_length... OK
  Applying auth.0009_alter_user_last_name_max_length... OK
  Applying auth.0010_alter_group_name_max_length... OK
  Applying auth.0011_update_proxy_permissions... OK
  Applying auth.0012_alter_user_first_name_max_length... OK
  Applying users.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying catalog.0001_initial... OK
  Applying catalog.0002_initial... OK
  Applying catalog.0003_course_catalog_cou_is_acti_d7f8f8_idx... OK
  Applying sessions.0001_initial... OK
test_generators_create_expected_data_and_are_repeatable ... ok
test_index_declared_on_model ... ok
```

```
test_index_exists_in_database ... ok
test_exists_count_values_annotate ... ok
test_n_plus_1_lecturer_before ... ok
test_n_plus_1_lessons_before ... ok
test_prefetch_related_lessons_after ... ok
test_select_related_lecturer_after ... ok
test_course_code_unique ... ok
test_course_created ... ok
test_lesson_relationship ... ok
test_custom_manager_active ... ok
test_default_role_is_student ... ok
test_enrollment_unique ... ok
test_lesson_order_unique_per_course ... ok
test_seed_demo_is_idempotent ... ok
test_select_related_uses_one_query ... ok
test_submission_relationship ... ok
Ran 18 tests in 7.198s
OK
```

3. Kesimpulan

select_related(). Mengambil 100 Course beserta Lecturer turun dari 101 menjadi 1 query karena Django memakai SQL JOIN. Waktu median turun dari 54.782 ms menjadi 2.632 ms (95.2%).

prefetch_related(). Mengambil 50 Course beserta Lesson turun dari 51 menjadi 2 query (1 query Course + 1 query seluruh Lesson, digabung di Python). Waktu median turun dari 42.319 ms menjadi 6.074 ms (85.6%).

Index. Index `(is_active, title)` mengubah rencana PostgreSQL dari Seq Scan menjadi Index Scan. Execution time pada EXPLAIN ANALYZE turun dari sekitar 0.568 ms menjadi 0.022 ms. Selisihnya kecil dalam angka mutlak karena tabel hanya 10.005 baris dan muat di memori; manfaatnya membesar seiring bertambahnya data. Index juga memakai ruang penyimpanan dan menambah biaya INSERT/UPDATE/DELETE.

Manfaat bagi LMS. Jumlah query tidak lagi naik seiring jumlah Course (1+N menjadi konstan), sehingga halaman daftar course dan materi lebih scalable. Seluruh 18 automated test tetap lulus setelah optimasi.

Semua angka di atas berasal dari output nyata command benchmark_week5 pada lingkungan pengerjaan; waktu pada komputer lain akan berbeda, sedangkan jumlah query dan perubahan jenis scan seharusnya sama.